Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · REPEATED-MEASURES ANOVA

3+ conditions on the same subjects

Table 1. Condition descriptives

Condition	N	M	SD
sleep_baseline	40	7.40	0.45
sleep_low	40	6.98	0.53
sleep_high	40	6.20	0.72

Table 2. Repeated-measures ANOVA

Source	SS	df	MS	F	p	partial η^2 [95% CI]
Condition	29.83	1.25	23.77	174.34	<.001	0.82 [0.76, 1.00]

Table 3. Sphericity (Mauchly's test)

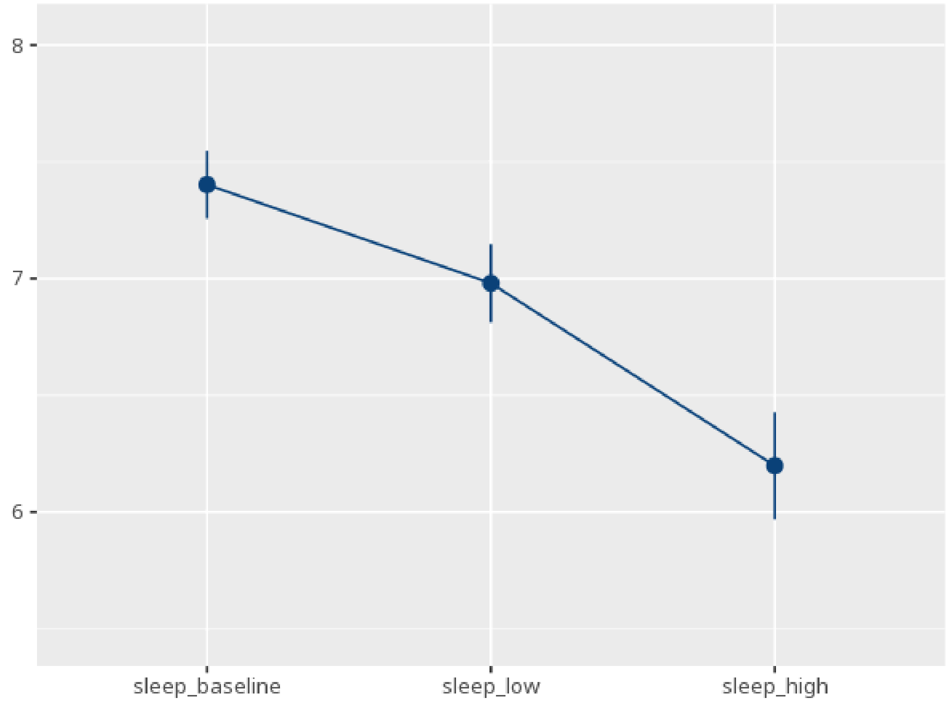
Effect	W	p	GG ε	HF ε
condition	0.41	<.001	0.63	0.64

when sphericity is violated the F-test uses the Greenhouse–Geisser / Huynh–Feldt correction; post-hoc table follows. Mauchly's test & the GG/HF corrections apply only when the repeated factor has 3+ levels (with 2 levels sphericity is automatically met and this table is omitted). - sphericity looks violated; check that a Greenhouse–Geisser or Huynh–Feldt correction is applied above

Table 4. Post-hoc comparisons

Pair	M_{diff}	SE	p_{adj}	95% CI
sleep_baseline - sleep_low	0.42	0.04	<.001	[0.32, 0.52]
sleep_baseline - sleep_high	1.20	0.09	<.001	[0.99, 1.42]
sleep_low - sleep_high	0.78	0.06	<.001	[0.62, 0.94]

Figure. Means across conditions



Understanding the numbers

Condition. Each row names one of the repeated measurements taken on the same subjects. Here, the descriptives table above reports 3 conditions, each with its own N, mean, and SD.

N. How many subjects contributed a value for that condition. Here, each of the 3 conditions above lists its own N in the descriptives table.

M. The average score for that condition. Here, each condition's mean is listed separately in the descriptives table above.

SD. How spread out scores are within that condition. Here, each condition's SD is listed separately in the descriptives table above.

Source. Names the within-subjects factor being tested - whether the average differs across conditions. Here, the tested source is Condition.

SS. The variability in the outcome attributable to differences between conditions. Here, SS = 29.83.

df. The degrees of freedom for the within-subjects effect, Greenhouse-Geisser/Huynh-Feldt-corrected when sphericity is violated. Here, df = 1.25.

MS. The effect's SS divided by its df. Here, MS = 23.77.

F. The ratio of between-condition to error variance - larger means conditions differ more than chance predicts. Here, F = 174.34.

p. The probability of a difference across conditions this large if the conditions truly had equal means. Here, $p < .001$ - below the $\alpha = 0.05$ threshold.

partial η^2. The proportion of variance explained by the within-subjects effect. Here, partial $\eta^2 = .82$ [.76, 1.00].

Effect (sphericity). Names which within-subjects effect Mauchly's test is checking for equal variances of the condition differences. Here, sphericity is checked for condition.

Mauchly's W. How far the pattern of condition-difference variances departs from the sphericity assumption (closer to 1 = more spherical). Here, W = 0.41.

GG ε. The Greenhouse-Geisser correction factor applied to the degrees of freedom when sphericity is violated. Here, GG ε = 0.63.

HF ε. The Huynh-Feldt correction factor - an alternative, usually less conservative, sphericity adjustment. Here, HF ε = 0.64.

M diff. The difference between two condition means in the post-hoc table. Here, each post-hoc row above reports its own mean difference between a pair of conditions.

SE. The standard error of that mean difference. Here, each contrast's SE is listed alongside its mean difference in the post-hoc table above.

p (adj). The pairwise p-value, adjusted for the number of comparisons made across conditions. Here, each pairwise comparison above carries its own multiple-comparison-adjusted p.

95% CI. The range likely to contain the true mean difference for that pair of conditions. Here, each contrast's 95% CI is shown in the post-hoc table above.

How to read this test

Tests whether the average differs across conditions measured on the same people. Check sphericity first - if violated, read the corrected F/p. A significant result means conditions differ; post-hoc tests show which. Your significance threshold (α) is 0.05.

APA template: A repeated-measures ANOVA (GG-corrected) gave $F(1.25,48.94)=174.34$, $p < .001$, partial $\eta^2=.82$ [.76, 1.00].

Statistical basis: Repeated-measures ANOVA (F-test for within-subjects conditions) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Mauchly's test of sphericity (rendered above the correction) (Mauchly, J. W. (1940). "Significance test for sphericity of a normal n-variate distribution." The Annals of Mathematical Statistics, 11(2), 204-209.); Greenhouse-Geisser correction (sphericity violated) (Greenhouse, S. W., Geisser, S. (1959). "On methods in the analysis of profile data." Psychometrika, 24(2), 95-112.); Huynh-Feldt correction (sphericity violated) (Huynh, H., Feldt, L. S. (1976). "Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs." Journal of Educational Statistics, 1(1), 69-82.); Estimated marginal means for post-hoc comparisons (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.). Full references in the exported CITATIONS.txt.